

Contact

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www.linkedin.com/in/rosaotavio
(LinkedIn)

github.com/otaviomrosa (Other)

Top Skills

Natural Language Processing (NLP)

Data Visualization

Statistical Data Analysis

Certifications

UR2PhD Undergraduate Research
Training Course Participant

Otavio Rosa

MS CS @ USF (CV & GenAI Research) → Targeting AI/ML
Engineering | PyTorch · LLMs · Computer Vision | Ex-SWE Intern @
ClearSet.AI
United States

Summary

MS AI @ USF | Graduating Spring 2027I build and study systems at the intersection of Computer Vision and Generative AI. My thesis research tackles a deceptively hard problem: can a neural network tell the difference between a real photo and an AI-generated one? Along the way I've done research on LLM bias, built real-time vision pipelines for a quadruped robot, and managed edge deployments on NVIDIA Jetson hardware. That mix of research depth and systems experience is what I bring to AI Engineering. Before the master's, I interned as a SWE at ClearSet.AI and co-organized HackUSF (200+ attendees) as Community Lead at Google Developer Group USF. Looking for AI/ML Engineering roles where research meets production. Stack: PyTorch · Computer Vision · Generative Models · LLMs · ROS 2 · Python

Experience

USF Bellini College of Artificial Intelligence, Cybersecurity and Computing

8 months

Research Assistant | Dr. Utkarsh Ojha

January 2026 - Present (6 months)

Tampa, FL

Conducting thesis research on generative image detection under Prof. Utkarsh Ojha, investigating the "Neural Turing Test" problem: can neural networks reliably distinguish real photographs from images synthesized by state-of-the-art generative models?

Research focuses on probing the representational limits of deep classifiers through alternative image modalities, exploring where and why discriminative ability breaks down between real and synthetic content.

Co-author on a paper submitted to NeurIPS.

Undergraduate Researcher | Language GRASP Lab

January 2026 - Present (6 months)

Tampa, FL

- Conducted large-scale behavioral analysis on open-source Large Language Models (LLMs) under Dr. Gene Kim to investigate nationality-based compliance bias. Developed methodologies for bias detection and mitigation utilizing mechanistic interpretability techniques, specifically neuron probing and integrated gradients.
- Presented original research findings, 'Quantifying Nationality-Based Compliance Bias in Qwen 2.5' at the 2026 AI+X Symposium.

Undergraduate Researcher | RARE Lab

November 2025 - April 2026 (6 months)

Tampa, FL

- Developed and optimized high-speed data acquisition pipelines for the Unitree Go2 quadruped robot, focusing on the intersection of human-robot interaction and computer vision.
- Engineered a synchronized Master-Slave automation system using Python and WebRTC to facilitate precise camera-projector calibration, significantly reducing the overhead for generating high-fidelity machine learning datasets.
- Managed edge-tier deployment on NVIDIA Jetson Orin Nano platforms, navigating complex networking challenges across ROS 2 middleware, SSH, and UDP/TCP protocols to ensure seamless communication between real-time motion controllers and high-level vision subsystems.

ClearSet.AI

Software Engineer Intern

May 2025 - July 2025 (3 months)

Tampa, FL

- Developed 30+ responsive and reusable UI components in Angular and TypeScript for core application features, which reduced new feature development time by an estimated 20%.
- Authored and maintained 30+ comprehensive SQL database scripts for schema migrations, data fixes, and new feature rollouts, ensuring zero-downtime deployments across 6 production releases..
- Identified and resolved 50+ bugs across the full application stack, from the C# .NET backend to the Angular frontend, directly improving system reliability and reducing the internal bug backlog by 25%.

Google Developer Group on Campus at USF

Community Lead

January 2025 - May 2025 (5 months)

Tampa, Florida, United States

- Co-led the planning and execution of HackUSF, a hackathon with 200+ participants and over \$3000 in prize, ensuring seamless logistics and resource coordination.
- Organized technology workshops by collaborating with 20+ industry professionals to enhance student learning experiences.
- Collaborated with local sponsors and university organizations to secure funding, resources, and mentorship support, reducing event costs by 40%.

Hillsborough College

Resident Assistant

August 2022 - May 2024 (1 year 10 months)

Tampa, FL

- Oversaw a diverse residential community of 400+ students, fostering community engagement through targeted programming and enforcing property regulations to ensure a safe and inclusive living environment.
- Mediated and resolved interpersonal conflicts, providing crisis management, documenting policy violations, and equipping residents with skills for future conflict resolution while liaising with HC Public Safety for appropriate escalations.

Education

University of South Florida

Bachelor's degree, Computer Science · (August 2024 - May 2026)

Hillsborough Community College

Associate of Arts - AA, Computer Science · (January 2022 - May 2024)